

patients with a median follow up of 48.5 months (IQR 13.6 – 134). Median survival for patients treated with radiation is 51.5 months (95% confidence interval (CI) 30.2-72.8 months) vs 62.2 months (95% CI 51.8-73.6 months) for patients treated without RT, with no significant difference in survival ($p=0.86$).

Conclusion: Despite multiple studies showing a local control benefit of radiation in the management of clival chordoma, this national hospital based analysis shows that radiation is broadly underutilized. Older age and treatment at an academic center were associated with treatment with radiation.

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Trends in Eye Plaque Brachytherapy for the Treatment of Choroidal Melanoma



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Purpose/Objective(s): Eye plaque brachytherapy (EPBT) is one of the primary treatment modalities for choroidal melanoma. Definitive treatment with EPBT is characterized by excellent local control rates, preservation of the eye. The Collaborative Ocular Melanoma Study group (COMS) studies established EPBT as an effective single modality treatment for small and medium sized lesions. However, single modality EPBT remains controversial for large lesions. Publications looking at single modality EPBT for large choroidal melanomas are exceedingly rare, and many predate significant advancements in EPBT technology, including improvements in surgical techniques, treatment planning systems, and dose delivery allowing dose escalation. The purpose of this study is to compare treatment outcomes for patients with choroidal melanoma treated with EPBT between earlier time periods and more recent time periods, to determine if recent advancements in the delivery of EPBT may have led to improved outcomes.

Materials/Methods: The National Cancer Database was queried (2004-2014) for patients with histologically-confirmed, localized choroidal melanoma treated with EPBT alone. Patients were divided into two cohorts based on year of diagnosis: patients diagnosed prior to 2010 and after. Patients were subsequently stratified by size (small, medium, large) based on COMS trial criteria. Kaplan-Meier analysis evaluated overall survival (OS). Significance of Kaplan-Meier estimates of OS was determined by log-rank testing.

Results: Results are shown in Table 1. A total of 5504 patients met inclusion criteria; 3223 (59%) patients were diagnosed prior to 2010 and 2281 (41%) after. 5-yr OS for all patients treated with EPBT alone prior to 2010 was 79%, compared to 82% after 2010 ($p=0.032$). On subset analysis, no statistically significant difference in 5-yr OS was found for patients treated before and after 2010 for either small or medium lesions ($p=0.989$, and $p=0.469$ respectively). However, for patients with large COMS lesions, 5-yr OS increased from 60% for patients diagnosed prior to 2010 to 76% diagnosed after ($p<0.0001$).

Conclusion: There is a statistically significant improvement in 5-yr OS survival for patients treated with single modality EPBT who were diagnosed after 2010 compared to before 2010. This change appears to be driven by improvements in 5-yr OS for large lesions. This change may be attributable to advancements in EPBT technology and/or changes in patient selection criteria. This study demonstrates that single modality EPBT

is a reasonable option for treating select patients with large lesions. Dedicated prospective studies are indicated to validate these results.

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Clinical Outcomes of Proton Beam Re-Irradiation for Recurrent Gliomas



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Purpose/Objective(s): Despite improved molecular subtyping, overall prognosis of recurrent gliomas remains poor. Salvage re-irradiation (reRT) for recurrence can result in significant local control and proton beam radiotherapy (PBRT) offers a potential advantage due to lower integral dose to previously irradiated tissue. There is limited data on PBRT reRT outcomes.

Materials/Methods: Twenty-two consecutive patients (median age 49, range 24-73) who received PBRT reRT for recurrent glioma between 2013 and 2018 were analyzed. The majority underwent PBRT reRT for recurrent Grade IV glioma (50%, 41% for WHO Grade III, and 9% for WHO Grade II). Prior to PBRT reRT, patients had a median of 1 surgery (range 1-4) and 2 systemic therapy regimens (range 1-5); 82% and 18% had 1 and 2 prior courses of RT, respectively, with a median duration of 27 months (range 7-232) prior to PBRT. Median total prior RT dose was 60 Gy (range 50.4-119.4). PBRT reRT was delivered using 3D conformal (50%), pencil beam (45%) scanning or a mix (5%). Planned reRT dose was a hypofractionated regimen in 45% of patients with doses of 30-40 Gy (RBE) in 10-15 fractions while the remaining 55% had doses of 45-60 Gy (RBE) in 25-33 fractions. Kaplan-Meier was used to determine local failure (LF), intracranial distant failure (DF), and overall survival (OS) from start of PBRT. Proportional hazards modeling was used to assess clinical predictors of these outcomes.

Results: The median OS post PBRT reRT was 11 months (6 months for Grade IV, 11 months for Grade III). Grade IV tumors had significantly poorer local control with a median time to LF after reRT of 6 months ($p=0.02$). Median time to DF was 7 months with no significant differences by WHO grade. Tumors with 1p19q codeletion ($HR=0.1$, $p=0.02$) and increased duration from prior RT ($HR=0.98$, $p=0.04$) were associated with improved OS post PBRT reRT while IDH1 mutant status trended toward better OS ($HR=0.3$, $p=0.06$). No molecular tumor markers were found to be associated with LF or DF, including MGMT methylation status. The number of prior lines of systemic therapy was also not associated with OS, LF, or DF. Patients had large reRT volumes with median PTV of 190 cc (range 3-676). PTV size was not prognostic of LC but hypofractionated courses trended towards poorer LC ($HR 4.2$, $p=0.08$) with no difference in DF ($p=0.97$) or OS ($p=0.12$). Two patients (10%) terminated PBRT early, one due to a mechanical fall and subsequent subdural hematoma; the other progressed during PBRT and required urgent debulking surgery. Three patients (14%) developed symptomatic radiation necrosis at 2, 2, and 12 months post reRT that were treated with bevacizumab and high dose steroids.

Conclusion: PBRT reRT was generally well tolerated with low rate of radiation necrosis and promising local disease control for recurrent glioma. While tumor grade and 1p19q codeletion status were predictive of OS, local control after reRT was associated with tumor grade and not tumor molecular markers. Further study is needed to elucidate which patients would most benefit from proton reRT.

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Abstract 2267; Table 1

COMS trial size	Diagnosed prior to 2010		Diagnosed after 2010		p-value
	Number of patients (%)	5 year OS	Number of patients (%)	5 year OS	
Small	707 (22%)	86%	480 (21%)	86%	0.989
Medium	2071 (64%)	80%	1449 (64%)	81%	0.469
Large	445 (14%)	60%	352 (15%)	76%	<0.0001
All patients	3223 (59%)	79%	2281 (41%)	82%	0.032